Q1. (2008) (22pts)

A hash table with the hash function h(k)=k mod m uses the double hashing strategy to resolve collisions. The <u>first three</u> keys 193, 297 and 321 produce the hash values 3, 12 and 17, respectively.

a) (5pts) What is the *prime* <u>hash table size</u>?

$$\left.\begin{array}{l} 193 \bmod m = 3 \\ 297 \bmod m = 12 \\ 321 \bmod m = 17 \end{array}\right\} \Rightarrow \begin{array}{l} (193-3) = km; k \in \mathrm{N} \\ (297-12) = lm; l \in \mathrm{N} \\ (2366-20) = nm; n \in \mathrm{N} \end{array} \quad \begin{array}{l} 190 = 2*5*19 \\ \Rightarrow 285 = 3*5*19 \Rightarrow m = 19 \\ 321 = 2^4 *19 \end{array}$$

(321-17)=nm 304=16*19

b) (10pts) The next key (i.e., the fourth key) that is hashed to the table <u>after three collisions</u> (each collision with a different key) is 326.

1. What is R in the double hash function?
2. Where is 326 hashed in the table?

In general R is taken as the greatest prime number less than m.

R=17

326 mod 19= 3

h(326,3)=[3+3(17-(326 mod 17))] mod 19 = [3+3*(17-3)]mod 19=45 mod 19 =7*

c) (5pts) Place, in respective order, the keys 107 and 127. In case of collisions use the double hashing strategy with the above R you found. Specify the number of possible collisions during hashing both keys!

h(107)=107 mod 19=12 ⇒ collision

h(k,i)=[h(k)+i*(R-(k mod R))] mod m

h(107,1)=[12+1(17-(107 mod 17))] mod 19=(12+12)mod 19=5⇒*
<u>107 is placed in #5 with 1 collision!</u>

h(127)=127 mod 19=13 ⇒ ***<u>127 is placed in #13 with no collisions!</u>***

Q2. (2013) (10+8+8=26 pts)

a) Insert the keys ***211, 376, 259, 127, 69, 84, 261, 273, 217*** into the hash table of length ***m=13*** using open addressing with the hash function ***$h_1(k)=k \bmod m$***. Use ***double hashing*** as the collision resolution strategy with ***$h_2(k)=R-(k \bmod R)$*** where *R* is determined by following our in-class discussion; calculate the collision frequencies (CF) into the table!

R=11 since *11* is the greatest prime less than *m*. Hence, $h_2(k)=11-(k \bmod 11)$

In general R is taken as the greatest prime number less than m.

CF	After\Hashtable	0	1	2	3	4	5	6	7	8	9	10	11	12
	Empty													
0	211				211									
0	376				211									376
1	259				211	259								376
0	127				211	259						127		376
1	69				211	259			69			127		376
0	84				211	259		84	69			127		376
0	261		261		211	259		84	69			127		376
0	273	273	261		211	259		84	69			127		376
0	217	273	261		211	259		84	69			127		376
9	X=118	273	261		211	259		84	69		217	127		376

b) (4+4 pts)

1. Find the average costs for insert and successful search for this table after all keys(including X) have been placed!
2. When would you decide to rehash? What would be the new table size (m_{new}) and *R* in the second hash function $h_2(k)$.

1.

Load factor: $a=10/13 \approx 0.77$

Cost for insert: $C_{ins}=1/(1-a) = 1/0.23=4.35$

Cost for successful search: $C_{search}=(1/a)\ln(1/(1-a)) = (13/10)*\ln(13/3) \approx 1.906$

2. I would start rehashing when a>1/2, i.e., after $\lceil \frac{13}{2} \rceil^{th}$ key (i.e., 7th key or 261) has been placed.

A typical example to the selection of the size of the new table is the first prime that is greater than two times the size of the current hash table.

13*2=26

Tablesize $m_{new}=29$,

In general R is taken as the greatest prime number less than m.

$R=23$.

Also acceptable is $m_{new}=31$, $R=29$ to have a closer R to m.

c) (HOMEWORK) Find a key, X, in the last line of the table above (i.e., after all keys are in part a are placed in the hash table) that can be placed in the hash table after ***9*** collisions.

h(k,i)=[h(k)+i*(R-(k mod R))] mod m

For a key to hash into the above table after 9 collisions,

1. the first attempt must be to hash it into 1 (i.e., X mod 13=1 $\Rightarrow$ X=13a+1, a$\in N^+$)

First attempt:0,1,3,4,6,7,9,10,12 (1st collision)

X mod 13 $\in$ {0,1,3,4,6,7,9,10,12}

h(k,1)=[h(k)+1*(11-(k mod 11))] mod 13 (2nd collision)
h(k,2)=[h(k)+2*(11-(k mod 11))] mod 13 (3rd collision)
h(k,3)=[h(k)+3*(11-(k mod 11))] mod 13 (4th collision)
h(k,4)=[h(k)+4*(11-(k mod 11))] mod 13 (5th collision)
h(k,5)=[h(k)+5*(11-(k mod 11))] mod 13 (6th collision)
…

h(k,8)=[h(k)+8*(11-(k mod 11))] mod 13 (9th collision)
h(k,9)=[h(k)+9*(11-(k mod 11))] mod 13 EMPTY

X mod 11 $\in$ {0,1,3,4,6,7,9,10,12}

2. the value of second hash function must be 3 (i.e., $h_2(X)=11-(X \bmod 11)=3 \Rightarrow$ X=11b+8, $\in N^+$).

All we ought to do is to find proper multiples of 11 and 13 so as to satisfy the given two equations. Trying several values we find that b=10, and a=9 are one pair of the values we are looking for. Hence ***X=118***.

Q3. (2014) (35pts) HOMEWORK

The first five keys of a seven-key sequence to place in a hash table are 345, 187, 141, 95 and 255. The hash function is of the form a_k ***mod m*** with *m* the size of the hash table. The strategy to resolve collisions is **double hashing**.

a) (5pts) What would be a good choice of *m* and *R*?

m=17, R=13

b) (15pts) Have the given five keys placed in the hash table such that the first four keys are hashed with no collisions while the last key (i.e., 255) can only be hashed after four collisions! Specify the hash value of the all five keys!

345 → 5
187 → 0
141 → 15
95→10
255 → 0,5,10,15,3

c) (5pts) What is the average cost for insertion considering the hash table in (b) ?

i. after the first four keys,

Cost for insert: $C_{ins}=1/(1-a)$ $17/13 \approx 1.31$

ii. When would you decide to rehash?

I would start rehashing when a>1/2, i.e., after $\lceil \frac{17}{2} \rceil^{th}$ key

after nine keys.

d) (10pts) Provided that the average cost for insertion must be less than 3/2, considering the hash table in (b),

i. how many keys at most can be inserted to the hash table?

$3/2=1/(1-a) \Rightarrow a=1/3 \Rightarrow$

a: load factor $n/17=1/3 \Rightarrow n=17/3 \approx 5.66$; **n= keys at most**

ii. How would you modify the hash table to make further insertions after that point?

A typical example to the selection of the size of the new table is the first prime that is greater than two times the size of the current hash table.

Rehashing with m=37 and R=31

Q4. (30 pts) (2007)

A set of keys is structured in a sequence of AVL trees as shown in Figure Q3. The ith node of a linked list contains a pointer that points to the root of the ith AVL tree and a key key_i that is less than the minimum key in the $i+1^{st}$ AVL tree and greater than the maximum key of the ith AVL tree. Hence, the keys in the list are in an ascending order.

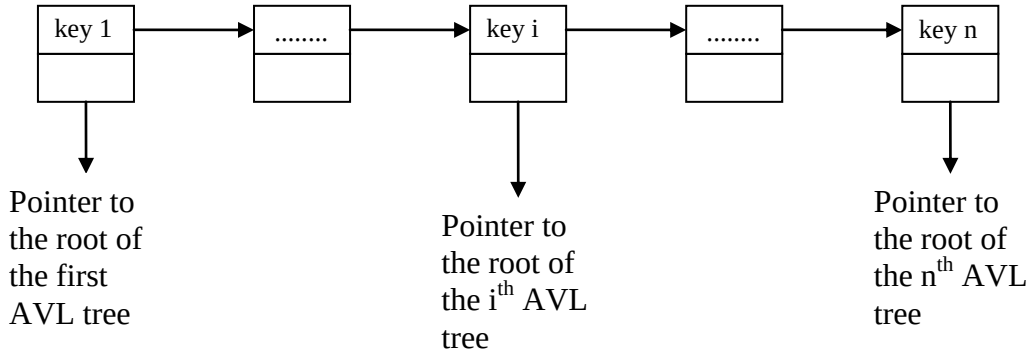


Figure Q.3: The data structure used in question 3.

a) (15pts)
Write an optimal search algorithm for the above data structure! Do not consider the modification part of the structure! Do NOT use the back of the paper for this part of question!

```
nodetype *search_LL (LL_p, int key) {
    nodetype *p; int EOLL=false;
    p=LL_p; if (p == NULL) EOLL=true;
    while ( !EOLL &&  p->key < key)
       {p=p->next; if (p==NULL)
EOLL=true;}
    if (!EOLL)
       if ( p->key > key ) //we found
where the key should be.
          search_AVL(p->rootptr,key);
       else return p; //p->key is the one
we are looking for.
}
search_AVL (nodetype *p, key) {
  if (p != NULL) {
    if (key < p->key)
       return search_AVL(p->left, key);
    if (key > p->key)
      return search_AVL(p->right, key);
    return p;
 }
}
```

b) (7.5pts)
Provide the asymptotic upper bound cost (i.e., execution time) of the above search algorithm in terms of *n* and *k* for a worst-case scenario! Show all your work!

$O(n*\log_2 k)$

Q5. (30pts) (2007)

a) (10pts)

How many 12-node binary tree topologies are there with the following two properties

- they are binary search trees (BST);
- their internal path length or total depth will be less than or equal to 39 or

$$d_{total} \leq 39$$

Show all your work!

Answer:

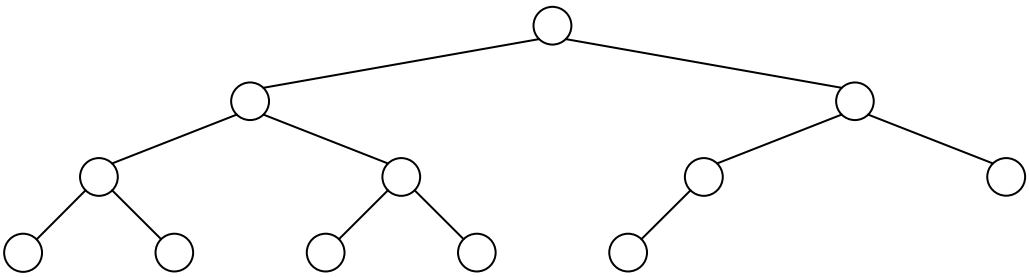
$$N = \binom{1}{1}\binom{2}{2}\left[\overbrace{\binom{4}{4}\binom{8}{5}}^{IPL=37} + \overbrace{\binom{4}{4}\binom{8}{4}\binom{8}{1} + \binom{4}{3}\binom{6}{5}\binom{10}{1}}^{IPL=38} + \overbrace{\binom{4}{4}\binom{8}{3}\binom{6}{2} + \binom{4}{3}\binom{6}{6}}^{IPL=39}\right] =$$

$$N = 7*8 + 70*8 + 4 + 56*15 + 4*6*10 = 1700$$

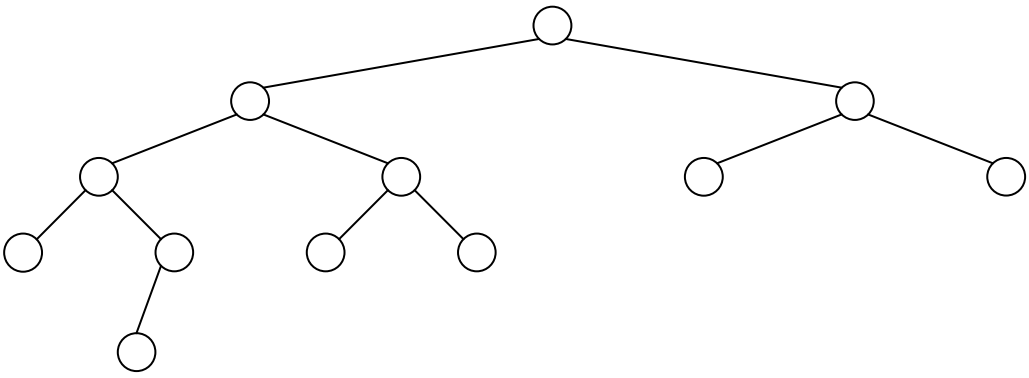
b) (10pts)
Show a sample topology for <u>each</u> class of BSTs in question (1.a)! Show all your work!

Hint: By sample topology I mean a BST topology with any combination of BST connections between nodes of consecutive depth levels as long as this sample topology has the same number of nodes at each depth level as the class of BST topologies which the sample topology represents.

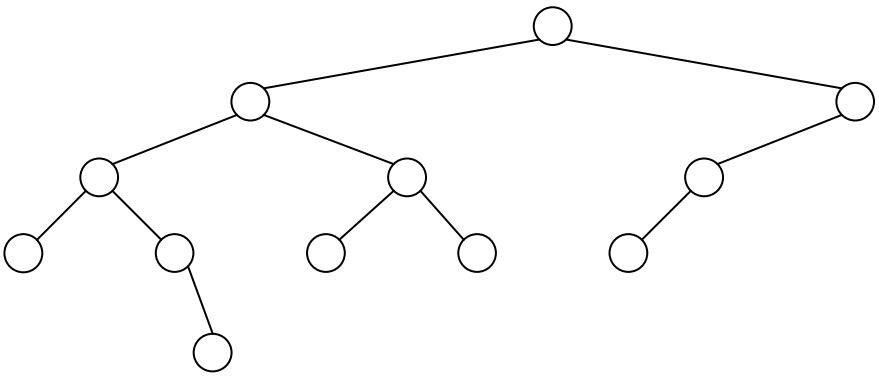
IPL=37.



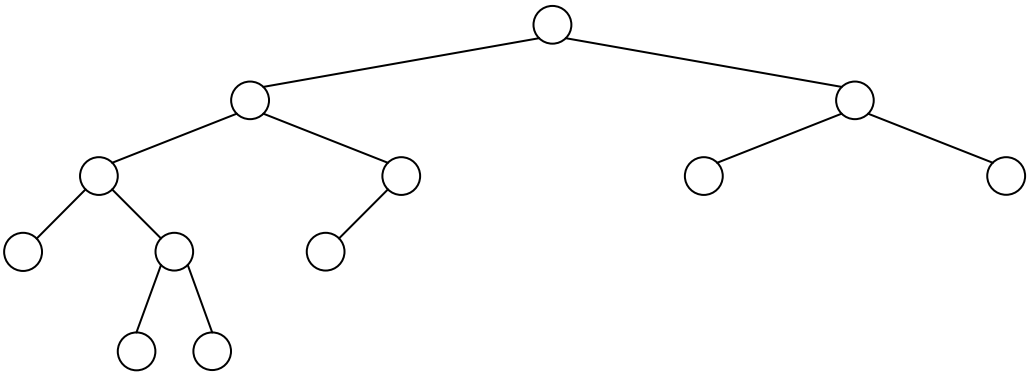
IPL=38



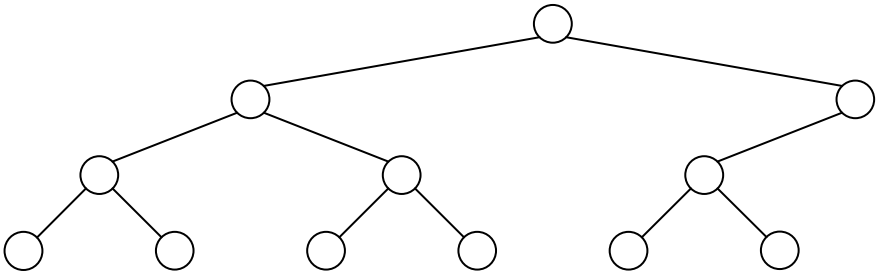
IPL=39



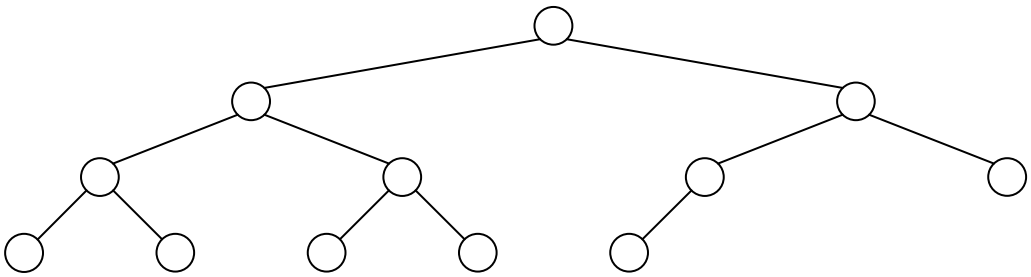
IPL=39



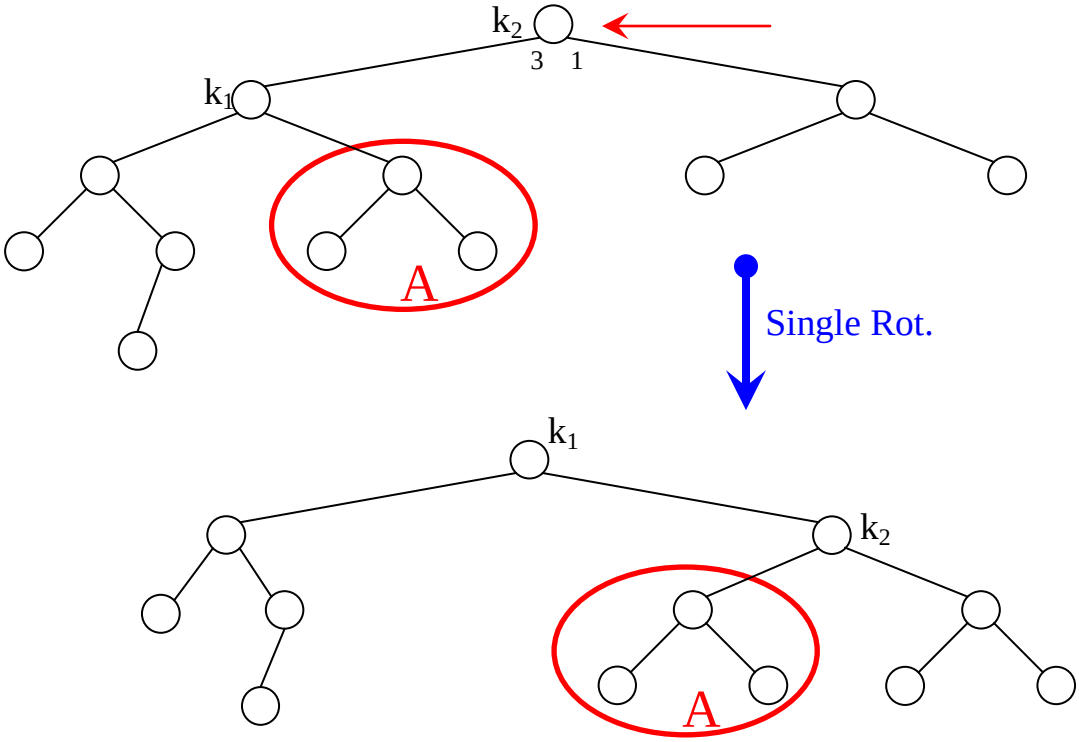
IPL=38



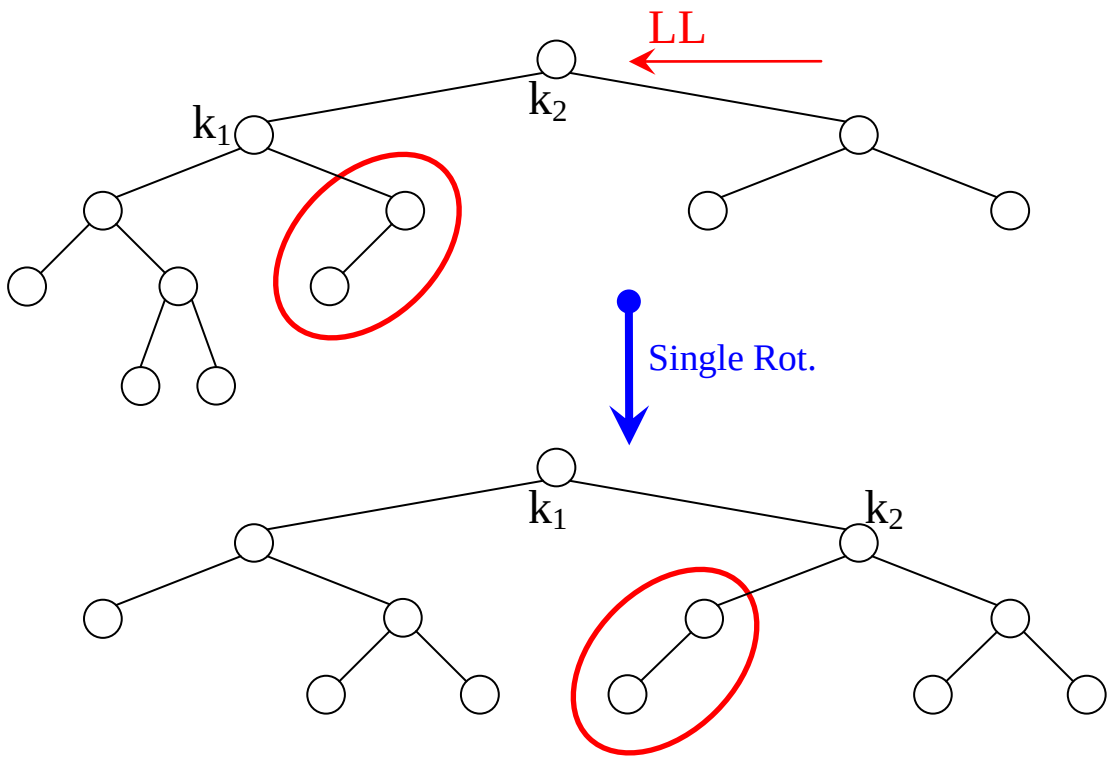
c) (10pts)
Consider all sample topologies in (1.b)! Restore the AVL tree property in all sample topologies where, you think, this property is violated. Show all your work!



This topology already possesses the AVL property.



The AVL property is restored.



The AVL property is restored.